

William Song

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EDUCATION

University of British Columbia

Sep 2023 – May 2027

Bachelor of Science, Major in Computer Science, GPA: 4.0

EXPERIENCE

Cloud Software Intern | Skydio

May 2026 – July 2026

- Designed and built an **AI Copilot** to assist operators and generate **X10 Drone** missions through natural language.
- Developed **MCP tools** for drone telemetry, mission control, geospatial reasoning, and trajectory analysis, with **Mapbox** and **Overpass API** integrations for terrain, world context, and infrastructure data.
- Prompt-engineered **Agent Skills** and tool call workflows, enabling the copilot to audit, explain, generate, and safely modify mission plans while ensuring **visible changes**, operator approval, and **sub-5 second** response times.
- Implemented a **multi-agent system** with specialized agents for both real-time and pre-planned drone operations.
- Deployed the **Vercel AI SDK** with **AWS Bedrock** providing **multimodal** and **streaming** support.

Software Engineer Intern | HeyGen

Sep 2025 – Dec 2025

- Co-developed an **AI Email-to-Video** pipeline integrated with **Amazon SES**. Built email processing and used **Temporal** workflows with **Redis** queues to orchestrate multi-stage video generation, including input validation, attachment parsing, context extraction, prompt assembly, inference, transcoding, and delivery.
- Implemented **usage-metered billing** for **HeyGen API** to support tiered, graduated, and overage-based payment plans, driving **30k+ /mo** in enterprise revenue with fullstack changes across **React**, **Flask**, **MySQL**, and **Stripe**.
- Created a **Slack App** that fetches context from DMs, channels, and threads to generate **AI videos** from messages.
- Used **PostHog** to run **A/B tests** on new upsell flows, improving add-on sales by **24%** through optimized **CTAs**.

Software Engineer Intern | Cyberium Group

Jan 2025 – April 2025

- Refactored a **NodeJS** ad recommendation backend into an N-tiered architecture, handling an **80%** increase in load, reducing crashes by **40%**, and applying a distributed deployment of individual layers for better horizontal scaling.
- Built wallet key management and integrated **Web3Auth** with OAuth2 for a **Webpack React** Chrome extension.
- Scaled a **Node Socket.IO** server architecture using an **Azure PubSub** server as the adapter to manage up to **1000** concurrent socket connections distributed across multiple **Azure VM** instances.
- Ensured infrastructure stability with **ArtilleryIO** load testing, validating reliability at **300 Socket events/second**.

Fullstack Developer | National Academy of Sciences Canada

June 2023 – Aug 2023

- Built a **WordPress** website for a non-profit organization to display news articles, publications, awards, and scientists.
- Used modular **Elementor** so non-technical users could maintain the site and trained **2** users to make content updates.
- Created a dashboard that tracked **400+** monthly page views and displayed article analytics.

Data Engineer Intern | Falcon Technologies

Sep 2022 – Dec 2022

- Aggregated licensee data from **8 provinces and 30 states** into a standardized **PostgreSQL** database.
- Scheduled **Python** and **Selenium** scripts to periodically **webscrape** new license information and update records.
- Maintained a database of **1 million** up-to-date license entries stored in a common searchable format.

PROJECTS

Macro Manager | NextJS, MongoDB, JWTs, PyTorch

Jul 2024 – Aug 2024

- Built an app to log meals and track nutrition, helping users monitor progress and reach dietary goals.
- Created many-to-many relationships in **MongoDB** to map references between **1000+** meals, ingredients, and recipes.
- Incorporated a custom AI model to help users find the nutritional information for a dish when provided with an image.
- Trained the CNN using **PyTorch** and the Food-101 dataset, achieving **81% top-1** and **95% top-3** accuracy.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, C, C++, SQL, PHP, Racket

Frameworks: ReactJS, NextJS, ExpressJS, PyTorch, Flask, Numpy, Selenium, WordPress, MaterialUI, TailwindCSS

Technologies: Git, Temporal, Vercel AI SDK, SocketIO, Stripe, PostHog, AWS, Azure, Docker, MongoDB, PostgreSQL